

HIIT and Its Influence on Stress, Fatigue, and Athlete Health

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Up to this point in the book, we've primarily been focused on the intricate methods involved in skinning the cat when it comes to high-intensity interval training (HIIT) programming for athletic performance. With any approach, however, it implies a linear relationship between the chosen training format and the performance outcome. That is, keep progressing the HIIT stimulus more and more, and performance capacity should follow. This is typically how we think, and we're as guilty as the next person in doing so at various times throughout our careers. Experience in the high-performance setting with coaches and athletes, however, has shown us that this relationship is not as simple as we may have wished. In this chapter, we welcome Philip Maffetone, a colleague with a vast amount of experience as a physician and coach, with involvement in virtually all athlete types ranging from endurance to motor sport, amounting to a 40 yr career as a clinician working with patients and athletes. Together, our aim in this chapter is to take a more holistic view of HIIT in outlining both the benefits and common pitfalls to such training within the context of its effect on stress, fatigue, and athlete health. HIIT has many benefits to performance as we've already outlined. Less discussed in the literature, however, are the potential negative consequences of the practice when things go wrong. As the old coaching saying goes, *speed kills*. This chapter deals

with aspects we don't always think about from the fitness standpoint, but need to if we want to prevent illness and injury, help athletes reach their human potential, and extend their performance lives. We must consider their health.

Health and Fitness Defined

The human body is generally quite resilient, as indicated by an athlete's ongoing and improving training and competitive cycles (74). However, injuries, poor health, reduced performance, and occasionally, and most unfortunately, even death, are realities. While we think of athletes and their amazing feats, we often automatically assume they have healthy bodies. However, that is not always the case. This apparent paradox is addressed by defining health and fitness separately (124):

- Health: a state of physical, biochemical, and mental-emotional well-being, in which all bodily systems (neuromuscular, hormonal, immune, digestive, etc.) function in harmony (figure 7.1).
- Fitness: the quality of human performance, the ability to compete at one's athletic potential.

Injuries, illness, and other mostly preventable but common conditions may be the best examples of

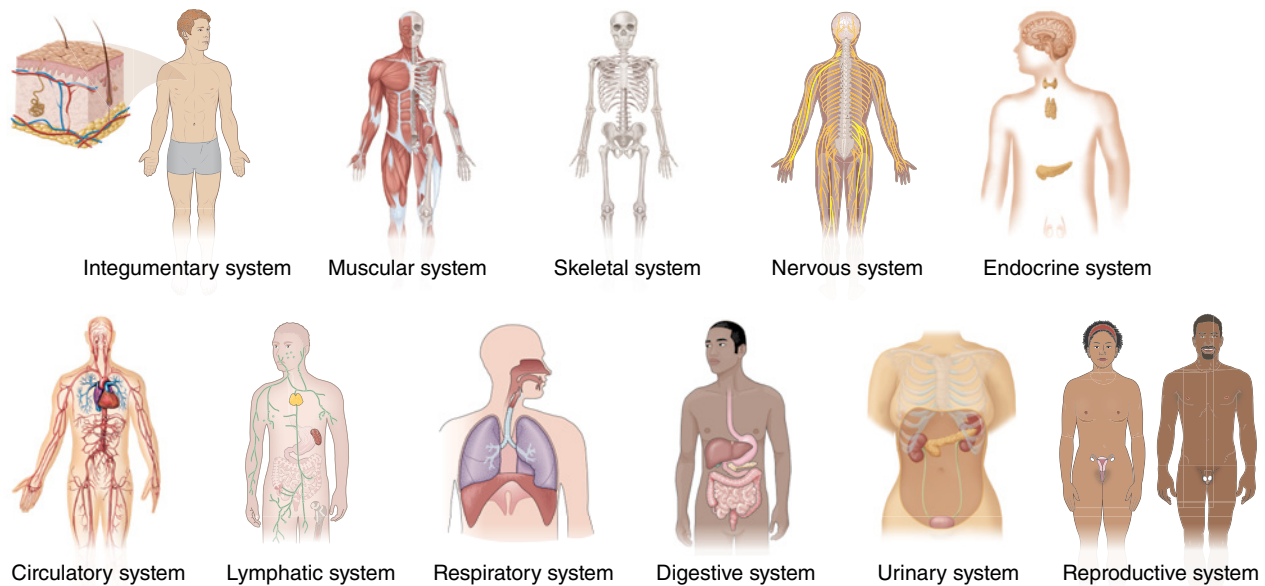


Figure 7.1 The 11 major physiological systems of the body that must be considered in the context of athlete health.

athletes who are fit but unhealthy. These conditions can also affect immediate and long-term training and competitive performance, the longevity of an athletic career, and current and future quality of life. By personalizing training and competition, and other lifestyle factors that impact the brain and body, athletes can be both healthy and fit (162).

Balanced Training

The many benefits of HIIT can be significant in helping athletes achieve optimal performances. The main physiological benefits, outlined in chapter 3, include improved aerobic oxidative, anaerobic glycolytic, and neuromuscular function. Additional benefits include enhanced bone remodeling, important for bone strength; support and function of muscles to enable human movement; and hormonal balance that includes the regulation of stress, sex, digestion, and other factors of importance for training, competition, and recovery.

Although the concept of balancing health and fitness in the name of preventing injury and illness to optimally progress performance ability is sometimes a difficult one to implement, a simple training equation may illustrate an important basic concept:

$$\text{Training} = \text{workout} + \text{recovery}$$

Training can be observed as the balance of an athlete's workouts plus recovery from them. Too much of the former, too little of the latter, or both,

can lead to excess training stress commonly called overtraining (104, 144).

In the previous equation, we can assume that training is the lifestyle an athlete chooses to live, which entails not only physical training but also the food consumed, stress management, a social life, and other daily routines including sleep. As such, overtraining refers to an imbalance of an athlete's lifestyle or a lifestyle the athlete is not adapted to (figure 7.2).

While HIIT provides adequate training stress to develop various health and fitness benefits, too much HIIT, either through volume, intensity, or lack of recovery, can add undue stress to the body. The topic of stress is an important one in relation to HIIT.

Stress

It is widely known that exercise offers many health and fitness benefits, the stimulus being a variety of catabolic and other physical and biochemical changes generally referred to as stress. More precisely, the anabolic adaptation mechanisms stimulated by sufficient training stress allow the body to benefit from HIIT. As described in chapter 3, proper adaptation to HIIT improves endurance, power output, or speed for a given metabolic cost, and ultimately performance.

Stress comes in many forms and can be categorized as physical, biochemical, and mental-emotional (153, 196). The molecular and physiological mechanisms of the various types of stress are well known and

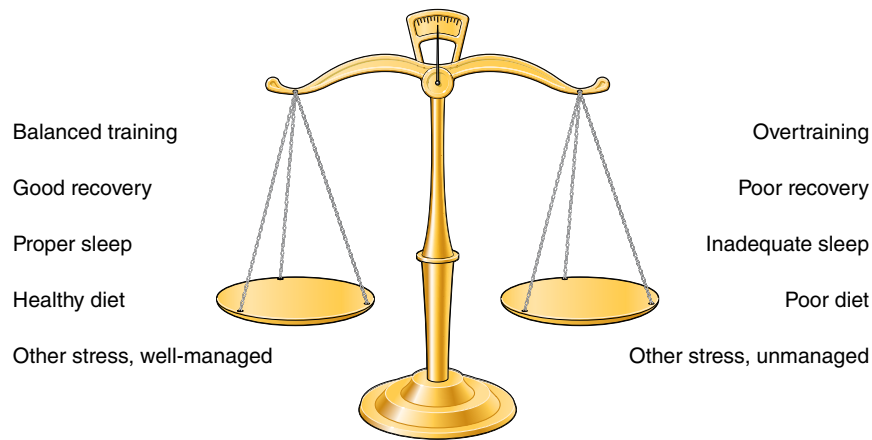


Figure 7.2 Athletes must balance stress in life.

applicable to all sports, and specifically to HIIT. Whether stress involves too much workout volume or intensity (the product of which is referred to as *training load*, see chapter 8), inadequate nutrition, or poor sleep or recovery, excess stress can impair health and ultimately fitness. Stress can be considered as the mechanism used by the body to adapt to the wear and tear of lifestyle. For the athlete, this means the accumulation of training and competition stress, with high-intensity activities typically producing the highest training loads (see chapter 8) and stress responses (195) (see chapter 9).

Defining Stress

Endocrinologist Hans Selye was the first to research biological stress and began demonstrating the existence of the brain-body adaptation mechanism and the hypothalamic-pituitary-adrenal (HPA) axis in the 1920s (197). He categorized the body's progressive response to all stress as the general adaptation syndrome, beginning with an acute stage, progressing to a second excess stress stage if recovery is not implemented, which can lead to a chronic and serious stage of stress. This general triad is outlined here because it is appropriate when describing athletic training, especially in relation to workloads such as HIIT, which can lead to fitness benefits with the potential for excess stress, e.g., overtraining.

In a sporting context, various terms are often thrown around to define the influence of stress associated with training and competition. These include burnout, staleness, failure adaptation, underrecovery, training stress syndrome, and unexplained underperformance syndrome, with the overtraining syndrome (OTS) the most commonly used term today

(103, 144). In addition, the general term *overreaching* is used in reference to increased training stress, with two specific components referred to as *functional overreaching* and *nonfunctional overreaching* (25). All these terms and conditions have an important common denominator—stress. To simplify these and other descriptions of stress, it may be best to consider the process as a spectrum (66, 123), one that also parallels Selye's three stress stages (described below).

HIIT Stress Benefits

An important feature of HIIT is the initiation of the first stage of acute stress. This allows the brain and body to adapt to an increased workload, and in conjunction with the process of recovery, various physical and biochemical adaptations are made that lead to improved competitive performance (see chapter 3). An individual's capacity to respond to this stress can vary, sometimes considerably (95), and occurs through the brain's hypothalamus and pituitary gland and the body's adrenal glands—the HPA axis.

Hypothalamic-Pituitary-Adrenal Axis

The HPA axis (figure 7.3) is a complex stress response mechanism important in the adaptation process to HIIT. A variety of neural and hormonal (neuro-endocrine) changes take place as a response to a high-intensity workout, such as increased heart and respiration rates, increased muscle activity and associated energetic processes, electrolyte and hydration adjustments, alongside discomfort and other processes. This response can even begin before the

The hypothalamic-pituitary adrenal (HPA) axis and the normal stress response to HIIT leading to adaptation.

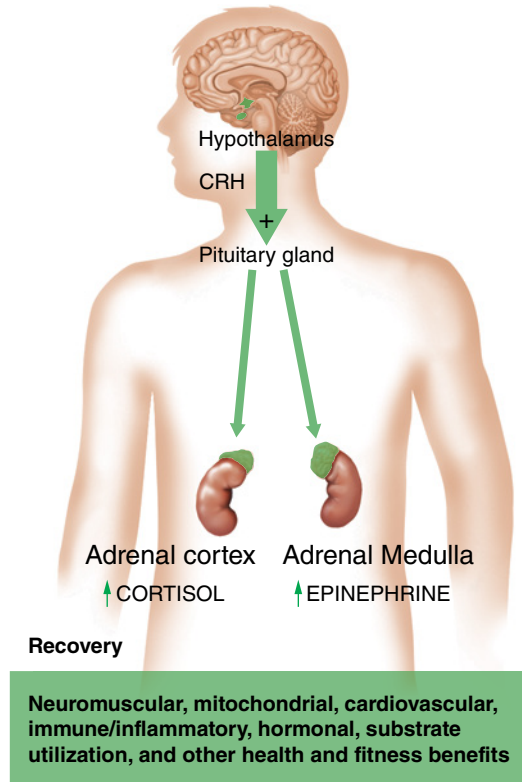


Figure 7.3 General overview of neuroendocrine and autonomic responses associated with healthy HIIT. HPA: hypothalamic-pituitary-adrenal; ANS: autonomic nervous system.

workout through pretraining anticipation (not unlike that of precompetition) sensed by the brain (40, 228). In addition to the hypothalamus, an area that influences autonomic function, the brain stem (helps control messages between brain and body) and limbic system (emotions) play important roles, too, along with body-wide hormone regulation via the pituitary. The hypothalamus also releases hormones (including corticotrophin-releasing hormone and arginine vasopressin) that act on the anterior pituitary to release adrenocorticotrophic hormone (ACTH), which in turn stimulates the adrenal gland's outer cortex to release various hormones that regulate glucose, electrolytes, and blood pressure, and primarily the stress hormone cortisol (which stimulates gluconeogenesis). Increased sympathetic activity releases another stress hormone, epinephrine, from the adrenal medulla. The resulting HPA actions impact the athlete's process of adaptation (with other axes also involved in this process, including hypothalamic-pituitary-gonadal and hypothalamic-pituitary-thyroid), with the inflammatory and immune mechanisms playing key roles.

Exercise is associated with a range of beneficial inflammatory and immune responses (135). HIIT, for example, can induce healthy levels of muscle and bone stress that trigger the process of repair, regeneration, and growth, moderated by inflammation (36). The process includes the release of a variety of proinflammatory cytokines, along with monocytes, neutrophils, and macrophages, along with antiinflammatory chemicals (135). The increased exercise intensity in particular also increases the release of oxygen free radicals (oxidative stress) by muscles (137). In addition to skeletal muscle, some of these actions, most notably the production of cytokine interleukin-6, infer benefits to cardiac muscle as well (138).

As outlined in chapter 3, adaptation to HIIT involves improving the function of mitochondria, the energy (ATP)-generating component of muscle cells, helping to manage oxygen free radicals (251). For example, healthy training adaptation upregulates antioxidant enzymes to cope with oxidative stress, especially in cardiac muscle (94). This process of recovery can last up to 48 h or more and possibly for days (224).

In summary, a healthy response to HIIT activates the neuroendocrine stress mechanisms originating in the HPA axis, resulting in the release of the adrenal cortex stress hormone cortisol and increased sympathetic activity releasing another stress hormone, epinephrine, from the adrenal medulla, with inflammatory and immune activity playing key roles in healthy adaptation. The physiological outcomes include various health and fitness benefits provided proper recovery takes place. Figure 7.3 overviews this mechanism.

While the acute benefits of HIIT can lead to adaptations that lead to improved performance (chapter 3), excessive stress from training and/or poor recovery can overwhelm the body's stress adaptation mechanisms leading to health impairments, including injury, illness, and reduced performance.

Athletes: Fit but Unhealthy?

In the pursuit to improve performance through higher training intensities or volumes beyond that which from the body can recover, athletes may inadvertently exceed their stress limits. This can result in various signs and symptoms—stress-related “injuries” that

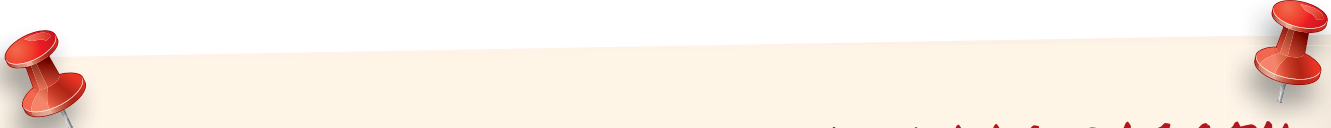
may be physical, biochemical, or mental-emotional. Some of these are listed in table 7.1. In addition to training, the accumulation of stress from an athlete's lifestyle, such as diet and social factors, can further lead to reduced fitness, as indicated by poor training and competitive performance. While traditional overtraining may be diagnosed *following* reduced performance lasting several months or performance *after* completion of a resting period of several days or weeks (77, 114), this does not allow for the potential of prevention, which is most valuable and preferred.

Assessing Training Stress

The recognition of excess training stress is an important step in preventing continued impairment, and is based on an athlete's current and past health and fitness history. *When* the transition from optimal HIIT to harmful stress occurs, if at all, varies considerably among athletes. Sometimes this appears as a physical injury or sudden onset of illness or extreme fatigue, often preceding diminished performance. Various subjective and objective physiological indications can help coaches and clinicians

Table 7.1 Potential Physical, Biochemical, and Mental-Emotional Signs and Symptoms Associated With Excess Stress in Athletes (103, 144)

Category	Signs and symptoms
Physical	Max and submax performance decrements Muscle soreness, stiffness, or fatigue General persistent fatigue Bradycardia Autonomic imbalance (including cardiac) Disease conditions such as asthma, thyroid disease, adrenal disease, diabetes mellitus or insipidus, iron deficiency with or without anemia, celiac sprue, hypertension
Biochemical	Excessive oxidative stress or damage Increased susceptibility to viral, bacterial, and other infections Inflammation Malnutrition Hormone imbalance Reduced muscle glycogen levels
Mental-emotional	Disordered eating Depression Insomnia Disrupted mood or behavior Loss of motivation Reduced mental concentration Anxiety



THE ATHLETE'S HEART AND SUDDEN DEATH

Physiologic adaptations that improve cardiac function are among the benefits of athletic training. They include increased stroke volume and cardiac output, longer diastolic filling time, and decreased oxygen demand, while maintaining normal systolic and diastolic function, with blood pressure and heart rate (HR) trending lower (see chapter 3). Training benefits, which include normal hypertrophic changes to the heart muscle walls, extend far beyond the cardiovascular system and are well known, with exercise far outweighing any health risks in most adults (68). Even extreme exercise training is well tolerated by healthy individuals, with increasing exercise volume and intensity further reducing cardiovascular risk (45). HIIT is recommended as a treatment strategy for patients with cardiovascular disease because it can provide benefits more rapidly than standard cardiac rehabilitation exercise (96).

Unfortunately, in the process of building fitness, athletes can also become unhealthy as discussed above, especially during high-intensity—high-stress—training (124). Combined with reduced health, high-intensity training and its associated increased HR and blood pressure and elevated stress hormones can contribute to sudden death in athletes with known or occult cardiovascular disease (45, 133).

Among the signs evident in unhealthy athletes is sudden death due to various cardiac injuries such as an acute myocardial infarction (heart attack) or arrhythmia (irregular electrical heart beat) (33). These individuals are often asymptomatic until during or within an hour of a cardiac event, and may be seven times more likely to die during activity than at rest (222).

The incidence of sudden cardiac death, the leading cause of death in athletes during exercise, has been estimated in young athletes <35 years of age at 1:917,000 athlete-years, however rates four to five times higher have recently been estimated, with men, African Americans, and male basketball players being at greatest risk (12).

A vital point about this important topic is that sport training in itself is not the cause of most sudden death in athletes, but acts as a trigger for cardiac arrest in the presence of underlying cardiovascular diseases predisposing to a cardiac event (45). In addition, sudden cardiac arrest does not always result in death, with many individuals surviving such an event, which may or may not occur during exercise.

In younger athletes (<35 years), the most common cause of sudden death is hypertrophic cardiomyopathy, accounting for approximately one-third or more of sudden deaths, with congenital coronary anomalies occurring in 15% to 20% of the cases and myocarditis (infection) much less common (45). In those >35 years of age, the most common cause of sudden death is atherosclerotic cardiovascular disease. As such, it is vital for clinicians to differentiate between normal hypertrophy developed from chronic training and its pathological state. Normal changes to an athlete's heart, which can imitate pathology, must be distinguished from serious cardiac disorders. Elite athletes are also not immune to cardiovascular disease (223), nor is it limited to athletes >35 years, and may account for 56% of the sudden deaths in young competitive athletes (132).

Cardiovascular causes for sudden cardiac death in athletes may be mostly preventative. For example, both genetic and lifestyle factors can contribute to pathologic hypertrophic cardiomyopathy, the latter being associated with cardiometabolic impairment including abnormal blood lipids and glucose, prediabetes and diabetes, and chronic inflammation. In athletes, hypertension is the most common risk factor for cardiovascular disease.

In athletes >35 years, the risk of sudden death due to ischemic heart disease increases progressively with age, where coronary artery disease accounts for more than half of the cases of sudden death (45), with atherosclerosis (a buildup of plaque in an artery) being a contributing factor. Coronary artery calcification is specifically associated with chronic high-intensity training (194), with vitamin D important in helping prevent calcification (21).

The prevalence of sudden death in athletes increased by 6% annually between study years 1980 to 1993 and 1994 to 2006, with 31% of the sudden death population occurring between 1980 and 1993 and 69% between 1994 and 2006 (132). As the data were largely limited to sudden deaths that became part of the public domain and records, the increasing number of fatal events observed over the study period may reflect enhanced public recognition caused by increased media attention. Corrado et al. (44) raised a reasonable concern about the reliability of the estimation of athletes' sudden cardiac death that could lead to an incorrectly low number of cardiac events and underestimation of mortality rates. Performance-enhancing drugs could also adversely affect heart health through cardiac toxicity, including anabolic-androgenic steroids, growth hormone, testosterone, erythropoietin (EPO), and others (33).

It is also possible that athletes experiencing fatal events previously had cardiac clinical screening that failed to detect a cardiac condition (180), with Maron et al. (132) estimating 30% of athletes cannot be identified reliably by preparticipation screening, even with ECG. Despite this, early identification of cardiovascular disease risks or cardiac abnormalities through a screening program could prevent sudden cardiac death, as recommended by the European Society of Cardiology, the American Heart Association, the International Olympic Committee, the American Society for Sports Medicine (64), and the American College of Cardiology (131).

Table 7.2 Important Assessment Tools Described in This Chapter

Assessment	Performed by
History and dialogue	Coach, clinician (athlete providing details)
Submaximal performance	Athlete (assessed by coach, clinician, see chapter 9)
Gait analysis	Coach, clinician, gait lab experts
Laboratory analyses (e.g., blood, urine, saliva)	Ordered by athlete, coach, clinician*
Complete clinical evaluation	Clinician
Heart rate variability (see chapter 9)	Athlete, coach, clinician
Dietary analysis	Athlete, coach, clinician
Urine color (hydration)	Athlete
Weight	Athlete, coach, clinician
Waist-to-height ratio	Athlete, coach, clinician

*Based on local regulations or laws.

monitor athletes before, during, and after HIIT, with a number of important assessment tools discussed in this chapter and listed in table 7.2. Chapter 8 focuses on the quantification of training stress for practitioners (termed training load), while chapter 9 examines methods of monitoring the athlete's response to training stress.

History and Dialogue

Sir William Osler, described as the father of modern medicine, said, "Never treat a stranger." Applied here,

the quote refers to the importance of knowing each athlete to help assess his or her readiness to start HIIT and whether the positive effects of HIIT stress turn negative. This may be most evident through a dialogue that uncovers various subjective symptoms. However, in our high-tech world (see chapters 8 and 9), reliance on digital devices often replaces some of the basics such as asking athletes about new or worsening symptoms. These subjective evaluations of wellness (see chapter 9) can be more important than high-tech measures (191, 218). These simple, cost-effective, subjective symptomatic evaluations

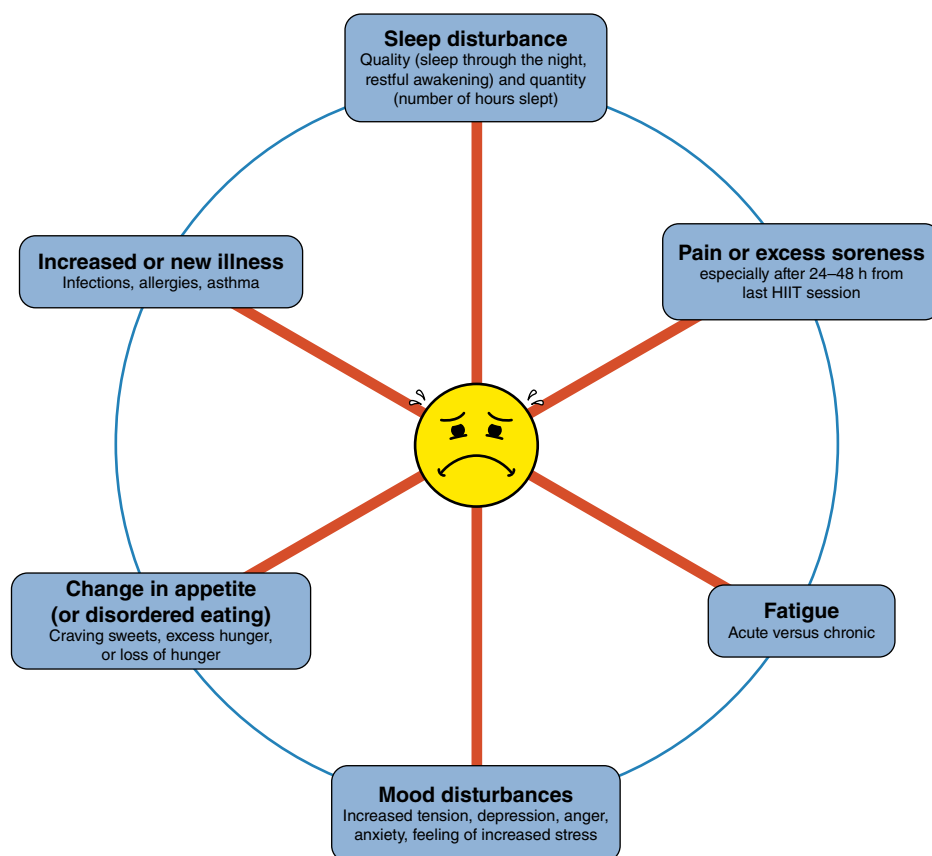


Figure 7.4 Signs and symptoms associated with the excess stress discussed in this chapter. Athletes can compare their current condition with that of “normal,” including worse or better than normal. This assessment can be performed weekly or monthly.

can be performed regularly, allowing a rapid adjustment to training schedules or other stress as indicated. The process can also help athletes better understand their own symptomatology and training progress. Examples of early symptoms include acute fatigue, increased illness such as infections, asthma or allergies, mood changes (associated with tension, depression, anger, vigor, fatigue, or confusion), and sleep disturbances (13, 191). Figure 7.4 lists some of these common early symptoms.

Assessing Responses to Submaximal Exercise

As expanded on in chapter 9, noninvasive heart rate assessment of submaximal exercise activity can accurately inform athletes, coaches, and clinicians about training status (25), with improvements in submaximal training HR occurring relatively quickly, sometimes within a week after the start of an endurance

training program (27). Submaximal testing can also reflect a state of overreaching (52, 202), and changes in the response to submaximal exercise may be one of the first objective indications of excess stress (114). In contrast with maximal tests, which are more fatiguing, submaximal intensity-based evaluations are more easily implemented during easy training periods, between HIIT days, and during the competitive season. For example, Buchheit et al. (26) showed that changes in submaximal exercise heart rate running at 9 km/h (5.6 mph) in a 5 min test were inversely related to changes in peak aerobic running speed (i.e., $V_{IncTest}$). Moreover, Le Meur et al. (113) have shown that heart rate recovery following submaximal exercise may be more discriminant than measures taken following maximal exercise for monitoring endurance athletes' responses to training stress.

Normally, periods of high-intensity training and/or increases in training volume, followed by adequate recovery, result in decreased resting, submaximal,

and maximal HR and improved competitive performance (114, 115). This can be evaluated by comparing a constant submaximal HR against run speed, cycling power, or other metrics. Expected training improvements are reflected in faster paces or speed (or increased power) at the same submaximal HR.

While reduced resting and submaximal HR can be considered a sign of improved fitness, measured by increased submaximal run speed, reductions in maximal HR can be reflective of improved competitive performance. However, without adequate recovery and/or excess training intensity or volume, athlete performance responses can be impaired, evidenced through either increased submaximal HR responses, reduced exercise run speed or power output, or both. Surveillance, in terms of the heart rate–exercise intensity response, can be used to monitor athlete progress and health, and is covered extensively in chapters 8 and 9.

Gait Analysis

The ongoing observation of gait can be a useful tool for evaluating excess stress. While even minor physical injuries are known to influence neuromuscular output during running, for example, and can influence gait (141), the accumulation of other physical, biochemical, or mental-emotional stress, possibly related to the intensity or volume of an HIIT session, can also adversely affect gait (67, 241). Fatigue can adversely affect gait through proprioceptive afferent feedback from muscles involved in motor control (146, 169). Specifically, fatigued muscles surrounding a joint can adversely affect the proprioception of that joint leading to potential abnormal movement (169, 207). Fatigue can also influence gait by slowing muscle reaction time (243). Gait analysis is a relatively simple process during on-field workouts and competition, with more in-depth details obtained in a gait analysis facility.

Clinical Assessment

When necessary, additional and more extensive assessments can be obtained by a health practitioner through a complete clinical evaluation based on the athlete's need. The justification for physical examinations, laboratory analysis, and other assessments are made by the clinician and therefore not detailed here. Clinical assessments may include a comprehensive health and fitness history and physical examination. In addition, basic laboratory evaluations

can include blood, urine, and hormone tests for immune, inflammatory, cardiometabolic, and other functions. More specifically, these may include:

- Blood tests such as a complete blood count and blood chemistry, including hematocrit and hemoglobin, liver function, glucose, and tests for electrolytes (Na, K, Ca, Mg), vitamin D, folate, iron and ferritin, along with C-reactive protein (a sign of excess inflammation).
- Urinalysis to help rule out infection, dehydration, and hematuria. It can also test for pregnancy.
- Salivary tests for cortisol or other hormones (98).

While there is no single test or group of tests specific for excess training stress, and most have limitations, a holistic approach that considers all aspects of an athlete's physical, biochemical, and mental-emotional condition is important because individual response to excess stress, and therefore signs, symptoms, and abnormal laboratory findings, can vary considerably (98).

Stages of Training Stress

A number of general stages of stress are outlined in this section with the heightened stage summarized in figure 7.5. The athlete response to stress, along with associated signs and symptoms, are based on the physiological progression of excess stress. Importantly, this is merely a broad-spectrum overview as athlete response to excess stress will vary considerably. For example, it is common for athletes to present with signs and symptoms in more than one stage of categories. Therefore, the following categories are provided only in the interest of offering general benchmarks within the stress severity continuum.

Acute Stress

The first stage of *acute stress*, sometimes referred to as overreaching, plays an important role in HIIT benefits, as discussed earlier (figure 7.3). However, excess stress can begin to affect the athlete's health and fitness in this stage via increased cortisol and sympathetic tone (87), producing abnormal signs and symptoms that may include:

- Reductions in sleep quality that can interfere with recovery, commonly observed as middle-of-the-night awakening, unable to quickly get back to sleep (187).

The hypothalamic-pituitary adrenal (HPA) axis and the heightened stress response leading to maladaptation.

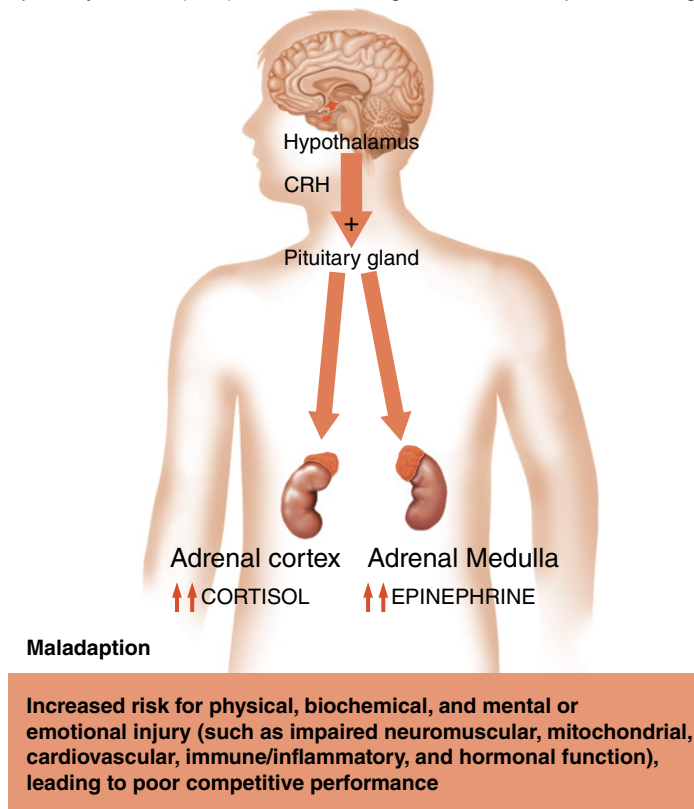


Figure 7.5 General overview of heightened neuroendocrine response to excess stress and maladaptation leading to impaired health and fitness. ACTH: adrenocorticotropic hormone; HPA: hypothalamic-pituitary-adrenal; ANS: autonomic nervous system.

- Feelings of depression or anxiety, fatigue, excess soreness, or pain, as the brain's executive functions are sensitive to excess stress. Other symptoms may be difficult for the athlete to precisely verbalize but could still be important (191, 218).
- While increased physical fitness is associated with cardiac vagal modulation that reduces resting and submax HR (26) and improves global heart rate variability (HRV) (176, 213), sympathetic overactivity can lead to *increased* resting HR and lowered global HRV (114). The regular use of a heart rate monitor can be an accurate way to help track HR changes throughout the year (25), the topic of which is expanded on in chapter 9.

With the acute nature of excess stress, the condition is often reversible within several days to weeks (106). Most important is to consider reducing training volume and/or intensity, such as using type 1 HIIT, with increased recovery during and/or between HIIT sessions. This is also an important time for coaches and athletes to consider other lifestyle stress

factors, especially diet as discussed below, to prevent progression of stress.

Advanced Stress Responses

The continuation of excess stress is sometimes called sympathetic overtraining, or nonfunctional overreaching, and is associated with increased ACTH from the pituitary with downstream increases in cortisol and epinephrine from the adrenal glands (196). This increased stress may be associated with sleep disturbances, daytime sleepiness, pain, and reduced desire to train. Even an acute bout of HIIT can elicit these neuroendocrine changes (120). The onset of immune (increased illness such as colds, flu and other infections, asthma) and GI dysfunction (from minor indigestion, excess gas or heartburn to vomiting or diarrhea during HIIT or competition) is also possible (51, 196).

Secondary to the now chronic cortisol elevations (figure 7.5), additional problems may include:

- Reductions in testosterone in men and women (102), potentially reducing muscle and bone

strength and sex drive, and further increasing fatigue.

- Reduced thyroid function may be the result of inhibition of the peripheral conversion of T4 (thyroxine) to T3 (triiodothyronine) (75). T3 may require up to 72 h following HIIT before returning to normal, with signs and symptoms of excess training stress similar to those of hypothyroidism.
- In female athletes, low energy, menstrual dysfunction, and reduced bone density, separately or in combination, make up the female triad, with the prevalence of at least one of these problems estimated at 60% (219). Disruption of the HPA axis can impair various other hormones including estradiol, testosterone, luteinizing hormone, and others, and cause menstrual dysfunction, such as amenorrhea and oligomenorrhea.
- Sleep disturbances may become more severe (117, 208) and are discussed separately.
- Neurological performance may become impaired, including reduced sensory skills such as keen awareness and fine hand-to-eye coordination required in sports such as soccer, baseball, basketball, racket sports, golf, and others (38, 139, 229). This may also include increased response or reaction time and increased number of mistakes in measures of attention, skills, and processing speed ability (54, 86).

Addressing these and other problems must involve attending to the primary cause(s). For example, prescribing hormone therapy or a sleeping aid may not address primary factors such as excess training stress, diet, or poor recovery.

Athletes with these increased levels of stress may be at risk for exercise-associated hyponatremia (EAH), usually defined by blood levels of sodium less than 135 mmol/L (160). EAH may develop in athletes during or within 24 h of longer and/or harder training or racing. However, low sodium may develop at any time during heightened levels of stress due to reduced aldosterone (secondary to high cortisol) and its influence on sodium retention, especially if combined with excess water intake (hyperhydration). In some athletes, water retention can result, leading to an increase in water weight after training or competition compared to weight before these events. EAH can produce symptoms of fatigue, mental disorientation, gait alterations, or breathing difficulty, or no symptoms at all, making it difficult

to diagnose and differentiate from dehydration. To differentiate, blood sodium levels can additionally be monitored during periods of high stress.

Recovery from these increased stress levels can take weeks to months depending on the severity of the athlete's condition. Temporary cessation of HIIT, reductions in training volume, and addressing other lifestyle stresses are important. Figure 7.5 is a general overview of the abnormal neuroendocrine response to excess stress.

Chronic Overtraining

The highest levels of chronic excess stress observed in clinical practice, referred to here as chronic overtraining, is a serious condition dominated by a significant autonomic dysfunction. It is associated with decreased hypothalamic and/or pituitary responsiveness to stress, with reduced growth hormone from the pituitary gland and low adrenal response to ACTH, sometimes called adrenal insufficiency (7, 24). The related adrenal insufficiency leads to abnormally low cortisol and other adrenal hormones, along with impaired hypothalamic-pituitary-gonadal axis function (24, 65).

A relative lack of sympathetic activity can result in an abnormally *decreased* resting heart rate, with continued worsening of both submaximal and competitive performance that can be a serious threat to the athlete's health (114). With their careers in jeopardy, these athletes usually require ongoing help from health practitioner(s) knowledgeable in sports medicine who can properly evaluate all physical, biochemical, and mental-emotional aspects of the patient. Returning an athlete to his or her former high-performance state may be difficult and could take months to years.

Remedies for the excess stress vary considerably with the athlete's particular needs. Some important factors to consider:

- A modification of the training schedule in stage 1 might include relying on lower-stress HIIT sessions (i.e., toward type 1), including shorter intervals and/or more rest between them, more rest or recovery days between HIIT workouts, or combinations of both.
- A *training taper* that includes up to a 50% reduction in training volume and intensity, for example, may further help speed recovery while increasing muscular strength, reducing the risk of injury and illness (119, 152, 155, 225).
- Excess stress associated with lifestyle must be addressed, including the diet.

- A complete clinical evaluation by a health practitioner may be necessary to help uncover physical, biochemical, or mental-emotional impairments associated with poor adaptation to HIIT.

Stress Recovery: Sleep

Good sleep quantity and quality can help athletes improve recovery from HIIT, increasing the opportunity for better training and competitive performance (76, 116, 204) with vital metabolic, immune, muscular, and other recovery processes occurring during specific stages of sleep (187). For example, Mah et al. (129) demonstrated that basketball players who obtained as much extra sleep as possible following 2 wk of normal sleep habits had faster sprint times and increased free-throw accuracy, along with increased vigor and decreased fatigue. Additionally, swimmers who increased their sleep time over their usual amount improved 15 m sprint, reaction time, turn time, and mood (128).

While adults may require a minimum of 7 h of quality sleep, with adolescents needing 8 to 10 h, these general guidelines could increase with the added stress of HIIT (187). As such, those who extend their normal sleep hours and “sleep in” may recover and perform better (116, 128, 129, 204). Unfortunately, the prevalence of poor sleep quality and/or quantity is common in athletes (187), who may be more sleep deprived than the general population (116, 184), with this problem being important feedback for coaches and clinicians (143).

Sleep *quantity* refers to total sleep time, while sleep *quality* refers more to factors such as time required for sleep onset, quality of rapid eye movement (REM) and non-REM sleep attained, waking or not during the night, and restfulness on awakening (161). Poor sleep quantity or quality can lead to:

- Symptoms of depression, tension, confusion, anger, decreased vigor, fatigue, and other mood states associated with neuroendocrine system disturbances (76).
- Both immune and inflammatory dysfunction (58, 61, 151).
- Enhanced daytime pain (which can interfere with nighttime sleep) (111).
- Impaired cardiometabolic factors (99, 210, 230), including poor glucose regulation, insulin resistance, and impaired appetite regulation, which may also increase hunger or cravings for sweets (209).
- Reduced estrogen and testosterone in men and women (50).

In addition to improving sleep quality and quantity, short (e.g. 30 min) naps can also help improve recovery (76). Longer daytime naps, however, may disturb nighttime sleep patterns.

Food, Stress, and Repair

The process of stress adaptation and proper recovery from training requires complex biochemical actions that continuously repair the body through processes

STRESS-RELATED FATIGUE

The normal process of recovery from HIIT is associated with mild to moderate fatigue and is considered a body-wide phenomenon that begins in the brain and affects the working muscles (159). Features of this neuromuscular fatigue include a temporary reduction in power output and submaximal performance of muscles, while increasing the perception of effort necessary to exert a specific force (57, 170). It can affect young healthy men and women who have no other signs or symptoms of dysfunction (243), affecting athletes from a wide variety of sports (4, 136, 207, 245).

Excess training stress can lead to excess fatigue, which can increase injury risk as muscles normally help disperse and share impact loads on joints and bones, with excess muscle fatigue reducing this protective action (182). Excess fatigue may be an acute response during or immediately after an HIIT session, with symptoms sometimes continuing up to the next HIIT session (238). This is an important issue to appreciate and relates to the programming aspects described throughout chapter 5, and the monitoring aspects discussed in chapters 8 and 9.

fueled by the consumption of food. Even a single HIIT session employs untold numbers of nutrients from food, including:

- **Macronutrients:** carbohydrate, fat, and protein; water; fiber.
- **Micronutrients:** all vitamins and minerals.
- **Phytonutrients:** thousands of plant-based compounds such as alpha- and beta-carotene, tocotrienols, and non-vitamin E tocopherols.

The American College of Sports Medicine and the International Society of Sports Nutrition recommend a “well-balanced diet” to meet all macro- and micronutrient requirements (30). However, nutritional deficiencies are not uncommon in athletes, even in those taking dietary supplements (6, 80, 105, 236). These nutritional problems can affect the HIIT response. For example, the very nature of the heightened metabolic requirement of HIIT increases oxygen free radicals that require various antioxidant nutrients from food to avoid unhealthy oxidative stress and poor recovery (216).

The term *diet* can be defined as the sum of energy (calories) and nutrients obtained from foods and beverages consumed regularly by individuals (8). A low-quality *poor* diet would include more processed, packaged, and preprepared foods that have lower nutrient density (such as micronutrients and fiber) versus natural, whole, unprocessed foods higher in nutrient density (43, 147). In addition, processed (refined) carbohydrates, including flour and sugar, foods containing added sugars, and sugar-sweetened beverages, substantially increase the glycemic index of meals and snacks and can contribute to chronic inflammation and other cardiometabolic impairments (206, 212). The name that encompasses these processed, low-quality items is *junk food*, sometimes called fast food, foods that are high in calories and low in nutrients (79).

The availability and affordability of junk food, including convenient or packaged items and those offered in restaurants, vending machines, cafeterias, and elsewhere, has led to an increased prevalence of poor diets, with sugar-sweetened beverages the single largest source of added sugar and the top source of energy intake in the U.S. diet (85). These products can increase hunger and cravings for more sweets (17). High-glycemic foods can also quickly impair fat oxidation rates (234, 240), contribute to the production of inflammation and pain (41, 92), and increase reactive oxygen species production (186).

Refined-carbohydrate foods are also associated with increased levels of insulin, a potent activator of the sympathetic nervous system (108).

Food intake and reward behavior is regulated by the hypothalamus and other areas of the brain (22, 181), especially in relation to the consumption of junk food (91). The combined HPA-axis stress (e.g., neuroendocrine imbalance) alongside the consumption of a high-sugar diet can promote inflammation, metabolic substrate imbalance (reduced fat oxidation), and maladaptation, and contribute to overtraining (figure 7.5) (124).

A major concern about the consumption of junk food, especially when regularly consumed, is that it replaces healthy food that could be providing many valuable nutrients to help recovery and adaptation from HIIT. Reading food labels, which list processed flour and sugar and carbohydrate content, helps make avoiding junk food easier. Natural whole foods are now sold by more stores and in some restaurants. Be aware of packaged products in natural food stores, which also contain many junk food ingredients. Table 7.3 provides examples of both junk food and alternative healthy food sources.

An important first step in assessing an athlete's food habits is performing a dietary analysis to evaluate diet quality. Some dietary analysis programs are available online and are free, such as the Cronometer (<https://cronometer.com>) and Senza (<https://senza.us>), and can provide reasonably accurate nutrient assessments, their primary purpose. However, some commercial programs also offer generic guidelines that may or may not be appropriate for a given athlete's individual needs.

Macronutrients

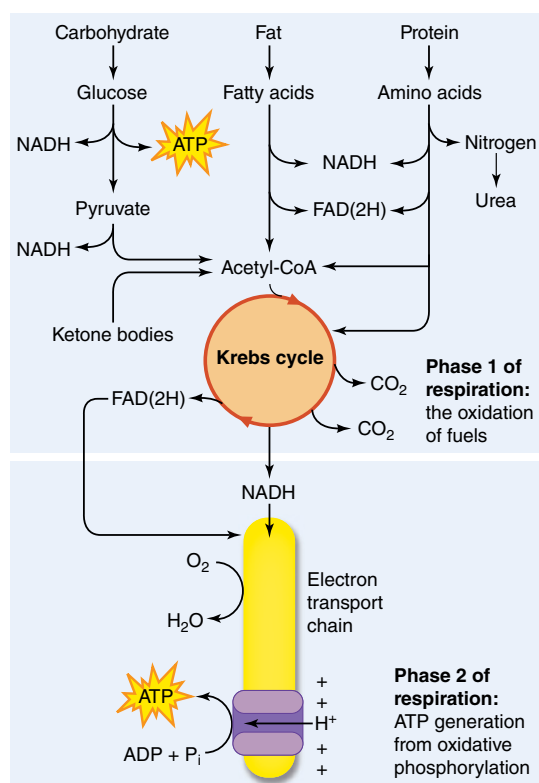
Once athletes eliminate junk food, balancing carbohydrates, fats, and proteins is easier. All three macronutrients are important in a healthy diet for athletes, in particular for those performing HIIT, as fatty acids (from fats) and amino acids (from proteins) are necessary for muscle recovery and repair through the regulation of inflammation and immune function. Balancing macronutrients involves individualizing food choices based on current needs, including eating *ad libitum*—when hungry rather than following a menu—a process humans have always done until relatively recently. Human research on self-selection of macronutrients associated with sports activity is limited, although it is known that the hypothalamus is associated with

Table 7.3 Common Junk Foods With Healthy Food Alternatives for Daily Meals and Snacks (43, 62, 147, 198)

Junk foods	Healthy alternatives
Prepared breakfast foods (cold cereal, instant oatmeal, pancakes, waffles, etc.)	Whole eggs (i.e., veggie omelet), unprocessed long-cooking whole grain cereals (i.e., oatmeal), fresh fruit and berries
Most bread (bagels, rice cakes, muffins, etc.), including most labeled "whole wheat"	100% whole grain bread products
Fast food (burgers, fishcakes, fries)	Vegetable salad, avocado, meat, fish, dairy
Processed lunch meat and fish (many brands have added sugar)	Whole pieces of meat, fish, or cheese
White rice, most pasta and noodles	Brown rice, lentils, beans
Most energy bars (high in sugar), soda, sports drinks, ^a sweetened protein shakes	Water, vegetable juice, coffee, and tea including herbal tea (without sugar)
Sweet snacks and desserts (cake, cookies, pie, sweetened yogurt, ice cream, candy, chocolate or sweetened milk, etc.)	Fresh fruit and vegetables, ^b unsweetened yogurt, raw nuts (almonds, cashews, macadamias)
Vegetable oils, trans fats (margarine and shortening)	Butter, olive oil, coconut oil

^aThe use of sports drinks around HIIT sessions and during competition could be considered a healthy choice relative to the goals of the session, as discussed in chapter 4.

^bHigh-glycemic vegetables include white potatoes and corn. High-glycemic fruits include bananas, grapes, watermelon, pineapple, and dried fruits (including raisins), and fruit juice.

**Figure 7.6** The role of food in the production of ATP.

the ability to choose foods based on the body's requirements (19).

Carbohydrate, fat, and protein are important contributors to the production of energy as ATP (chapter 3) and require the actions of micronutrients (see figure 7.6).

Traditional macronutrients for total daily energy in adults, set by the Food and Nutrition Board of the Institute of Medicine (130) is as follows:

- Carbohydrate at 45% to 65%
- Fat at 20% to 35%
- Protein at 10% to 35%.

However, these recommendations are a general guideline and inherently subjective as they don't take individual needs into account. For example, some athletes may perform best with significantly different macronutrient percentages, as discussed below.

Dietary Carbohydrate

Some traditional carbohydrate recommendations for athletes performing high levels of training are 8-12 g/kg per day (29). However, this estimation does not necessarily apply to everyone, nor is it followed. In a study of elite athletes (327 men and 226 women) across a wide range of sports including

endurance, team, and strength-focused, normal macronutrient intake was evaluated, with more than half of the individuals consuming <5.0 g/kg of carbohydrate per day (236). As dietary carbohydrate is technically not an essential nutrient, with glycerol (from fats) and a variety of amino acids (from protein) capable of converting to glucose (242), there is a continued interest in lower carbohydrate and higher fat and protein in sports, which challenges traditional recommendations (16, 156, 166, 249). Examples of good sources of carbohydrate are lentil stew with vegetables and fresh fruit and berries.

While there is no official definition of low- or very low-carbohydrate diets, three commonly used terms associated with reduced carbohydrate diets are as follows (59, 158).

Moderate-carb: 26% to 40% of daily meals, with 140 to 210 g of natural carbohydrate per day; fat content is 100 to 130 g or 43% to 57%.

Low-carb: 11% to 25% of daily meals, with 51 to 139 g of natural carbohydrate per day; fat content is 135 to 165 g or 58% to 71%. Low-carbohydrate intake may begin producing measurable amounts of ketone bodies (assessed through blood, urine, and breath) that are used for energy, and may be important health-promoting signaling molecules (233).

Very low-carb: 5% to 10% of daily meals, with 25 to 50 g of natural carbohydrate; fat content is 168 to 182 g or 72% to 78%. Very low-carbohydrate intake leads to higher levels of ketone bodies (beta-hydroxybutyrate concentration >0.5 mmol/L), referred to as *ketogenic*, with individuals in a state of nutritional ketosis.

Various other traditional definitions are sometimes used (8).

Extending further from the information outlined initially for us in chapter 4 (carbohydrate training low), some benefits of reducing carbohydrate foods from traditionally recommended levels include:

- **HIIT and competitive activities.** Increased fat oxidation and glycogen-sparing effects that can help maintain and/or improve power output during repeated sprints for athletes in team sports such as soccer, rugby, and basketball (35), gymnastics (167), and endurance athletes (109, 234, 240).
- **Weight and body fat.** Long-term low-carbohydrate, high-fat eating has been more or

as effective as high-carbohydrate, low-fat eating in reducing body weight and fat mass (28). This is a healthy alternative for athletes in weight-sensitive sports, who, through unhealthy strategies of energy restriction, dehydration, and other dangerous methods to reduce weight, can impair both performance and health (97).

- **Health.** As reviewed by Chang et al. (35), many studies have shown that long-term lower-carbohydrate intake is not harmful provided there is sufficient energy and protein intake and can actually reduce the risk of increased body fat, metabolic imbalances, cardiovascular disease, diabetes, and other conditions.

These changes can occur following metabolic adaptation to a lower-carbohydrate, higher-fat eating routine, which may require at least 2 to 4 wk. Acute withdrawal from a high-carbohydrate diet, however, can temporarily reduce muscle glycogen and blood glucose, increase fatigue, and impair performance (35, 109).

A case study of a professional female triathlete of 13 years demonstrated how the process of individualizing macronutrient intake, resulting in very low-carbohydrate eating, can be accomplished (125). Based on signs and symptoms, including extreme hunger and fatigue, increased need for nutrients (food and drink) during training, menstrual stress (painful and irregular periods), and severe GI distress, especially during races, alongside laboratory evaluations such as elevated fasting glucose, the following dietary recommendations were made:

- eliminate refined carbohydrates,
- reduce natural carbohydrate, and
- make up the caloric difference by increasing dietary fat.

Reducing carbohydrate foods and monitoring symptoms over a period of about 2 to 4 wk resulted in reductions in hunger, fatigue, food needs during training, and GI distress, with the ultimate carbohydrate content decreasing from 73% (475 g) to 12% (78 g) of total calories, whereas fat content increased from 14% (40 g) to 75% (217 g) and protein levels remained constant at 13% (85 g) (see figure 7.7). Within 2 mo, the athlete also reported increased perception of daily energy during and between training sessions, normal menstruation, and reduced need for daytime naps, with normalization of fasting

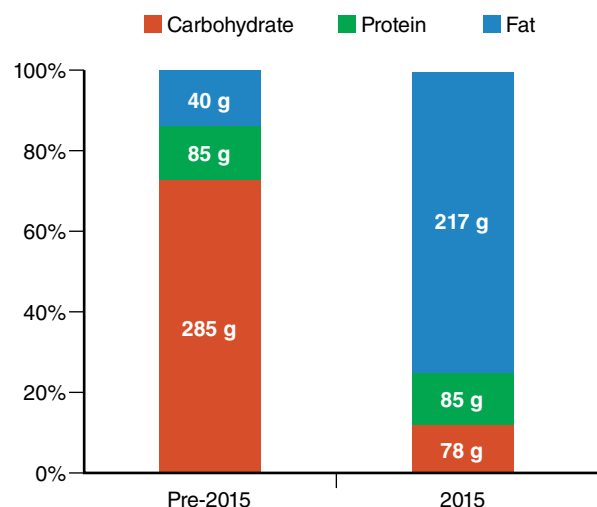


Figure 7.7 Daily macronutrient intake before and after dietary shift and intervention.

Reprinted by permission from P.B. Maffetone and P.B. Laursen, "Reductions in Training Load and Dietary Carbohydrates Help Restore Health and Improve Performance in an Ironman Triathlete," *International Journal of Sports Science & Coaching* 12 no. 4 (2017): 514-519.

glucose. In addition, cycling power output increased by 20 W and run pace increased (12-15 s/km) at the same training heart rate (141 beat/min), and, with the exception of water, nutrition was no longer required during long rides (~4 h). Race calories consumed (based on energy needs) were reduced from ~400 kcal/h in previous years to ~175, 145, and 130 kcal/h over the course of the next three Ironman events. Signs and symptoms of GI stress during competition ceased to occur, and 2 of 3 races were personal best times, with the final event completed in 8 h, 52 min. The levels of carbohydrate were well below typical recommendations (29), with fat intake higher than the ~30% considered appropriate, and ~85 g of protein for this 60 kg (132 lb) athlete, which is within some common recommendations ranging from 1.2 to 2.0 g/kg (174), although Paoli et al. (167) have recommended 1.3 to 2.5 g/kg of protein to ensure the maintenance of muscle mass, gluconeogenesis, and fat oxidation when consuming low-carbohydrate, high-fat diets. While maintaining very low-carbohydrate, high-fat intake, all markers of health and fitness improved, both short and long term.

Dietary Fat

Natural, healthy fats (also called oils) in the diet are necessary to help athletes recover from HIIT and improve overall health. Examples of healthy fats include olive and coconut oils, butter, the fats in

meats, eggs, and dairy, and other foods high in healthy fats such as avocados and raw nuts such as macadamias, almonds, and cashews. These fats can be used by the body for beta-oxidation to provide energy and regulation of inflammation (discussed below). Unhealthy fats include margarine and shortening that contain trans fats, and vegetable oils high in omega-6 fats (corn, peanut, safflower, soy, canola) (93), which can promote inflammation under certain conditions.

After consumption, dietary fat is usually deposited in fat stores. Even the leanest athletes when healthy can use this fat to supply a substantial portion of their total 24 h energy needs and training energy, including for HIIT (48, 81, 183, 233, 249). Generally, the combined amounts of fat and glucose used for energy during training vary with intensity, with increased fat oxidation occurring at lower intensities and lower levels as intensity increases. In addition to movement, fat oxidation provides energy for slow-twitch aerobic muscle fiber function, important to help support bones and joints, vascularization, and antioxidant activity (175).

Fats also help regulate inflammation through the actions of essential fatty acids: omega-6 (containing linoleic and arachidonic acids) and omega-3 (containing linolenic and eicosapentaenoic acids) (20, 93, 247).

Omega-6 oil contains the essential fat linoleic acid (LA), which normally converts to anti-inflammatory chemicals (collectively called eicosanoids). However, poor eating can divert LA to inflammatory chemicals instead through two key mechanisms. Excess intake of LA-containing omega-6 vegetable oils (corn, peanut, safflower, soy, canola), which has doubled in the past 100 years in Western countries (93), especially in junk foods, can convert LA to inflammatory chemicals. Carbohydrate-stimulated insulin (especially high-glycemic foods) further promotes the conversion of LA to inflammatory chemicals. In addition, a poor diet—one low in micronutrients and/or protein and high in trans fats—along with excess stress, can reduce production of anti-inflammatory chemicals from LA. Arachidonic acid (AA) is also an essential fatty acid that normally produces appropriate amounts of inflammatory chemicals (important in recovery from HIIT). The most common dietary source of AA is chicken, with other meat, egg yolk, and dairy containing lesser amounts. However, excess intake of omega-6 fats, which convert to inflammatory chemicals, do so by increasing AA. Beware of the dietary supplements containing primrose, borage, or black currant seed oils (concentrated omega-6 oils containing high levels

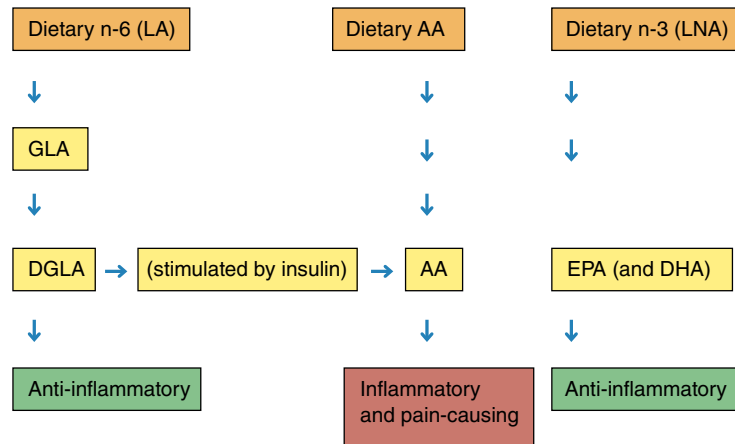


Figure 7.8 The relationship between dietary fats, refined carbohydrates, and inflammation. Excess omega-6 fats (corn, canola, soy, peanut, and safflower oils) and excess sugar or refined carbohydrate-stimulated insulin can drive omega-6 fats away from their potential anti-inflammatory pathways to inflammation through conversion to arachidonic acid. Some foods, especially chicken, also contain arachidonic acid. A different pathway containing omega-3 fats, primarily EPA-containing fish and fish oil supplements, can increase anti-inflammatory chemicals (23, 93). n-6: omega-6 fats; LA: linoleic acid; GLA: gamma linolenic acid; DGLA: dihomo-gamma linolenic acid; AA: arachidonic acid; n-3: omega-3 fats; LNA: alpha linolenic acid; EPA: eicosapentaenoic acid; DHA: docosahexaenoic acid.

of gamma linolenic acid), which can convert to AA in the inflammatory pathway.

Omega-3 fats, common in coldwater fish and seafood, with smaller amounts in beans and flaxseeds, contain the essential fat alpha linolenic acid (LNA) and also play an important role in producing anti-inflammatory chemicals. LNA must first be converted to eicosapentaenoic acid (EPA) (along with docosahexaenoic acid) for anti-inflammatory actions (23), a process requiring micronutrients and protein, but is impaired by trans fat and excess stress. However, in humans, plant-based omega-3 oils (such as from flaxseeds and beans) are poorly converted to EPA. Animal foods, primarily fish, already contain EPA. Coldwater fish such as salmon, anchovies, herring, and sardines have the highest food sources of EPA.

Figure 7.8 shows the relationship between dietary fats, their respective fatty acids, and the production of inflammatory and anti-inflammatory chemicals. Inflammation is also associated with pain.

In summary, the combination of a junk food diet and excess training stress can increase the risk of chronic inflammation (41, 92), which can trigger physical, biochemical, and mental-emotional injuries, including pain. Associated suppression of the immune system also increases the risk of infections, allergies, and asthma (215). These inflammatory and immune responses are comparable to those created by serious trauma and infections, and involve complex mecha-

nisms described elsewhere (100). Figure 7.9 is an overview of this diet and training paradigm.

Dietary Protein

High-quality healthy protein foods include *unprocessed* meat such as beef, pork, lamb, and fish (62), along with eggs and dairy (221). During periods of reduced protein foods (such as during travel), dietary supplements, including healthy protein drinks, may be useful to help avoid lower protein intakes. Reading nutritional labels helps avoid added sugars, common in these foods.

The intake of adequate dietary protein can help ensure athletes recover by enhancing skeletal muscle, bone and tendon adaptation, along with various metabolic benefits (72, 221). Protein may also supply up to 15% of an athlete's energy needs (63).

Protein needs may be more individualized than by that predicated by sport, with general requirements similar between athletes engaged in aerobic and sprint activities and most other sports, although higher intakes are indicated during periods of high-intensity training (221). For example, while the recommended daily allowance (RDA) of protein is 0.8 g/kg, consumption at double this level (1.6 g/kg) or higher can help preserve lean muscle and reduce excess body fat (8, 15, 112). Longland et al. (118) recently showed that during periods of high-intensity interval sprints and resistance training,

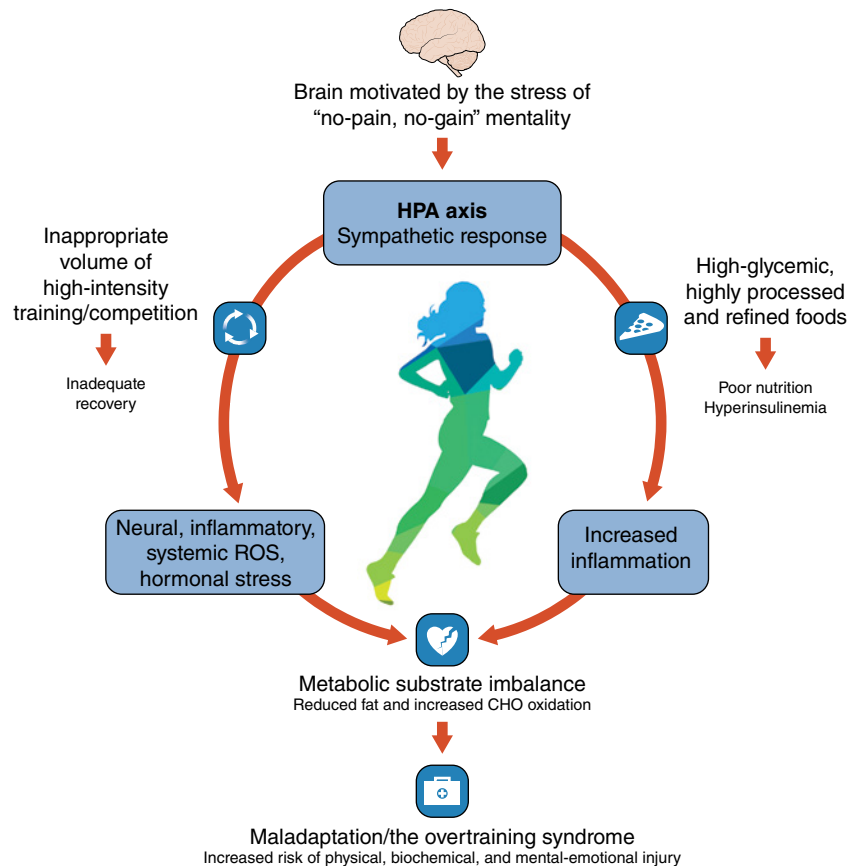


Figure 7.9 The fit but unhealthy eating and training paradigm.

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protein intake at 2.4 g/kg caused lean muscle gains (1.2 kg) and fat loss (4.8 kg), while 1.2 g/kg preserved lean mass and fat loss. Instead of daily recommendations, muscle adaptation to training may be maximized by ingesting ~0.3 g/kg body weight following high-intensity training (221). In addition to post-training intake, spreading protein intake relatively evenly throughout the day, such as in 3 or 4 meals or every 3 to 5 h, may help improve digestion, absorption, and utilization (174).

Water

As a macronutrient, water is obviously important for proper hydration in all aspects of health and performance. We are made up of more than 50% water. Daily water needs are satisfied by drinking plain water plus the water content in foods and other beverages. Science is without a consensus on drinking strategies for athletes, as there is likely a range of successful approaches that reflect individual needs (10, 56, 70) and include:

- A fluid plan that follows a certain volume per hour based on the sport, exercise duration, and outside temperature.
- Ad libitum drinking, e.g., whenever and in whatever volume an athlete chooses.
- Drinking to thirst, very similar to ad libitum, and equally effective (47).

As we touched on in chapter 4, increased muscle activity during HIIT increases heat production that is compensated for through sweating, which requires adequate water intake to prevent excess body temperatures (192). Elevations in core body temperature for prolonged periods of time that exceed 40°C (104°F) can increase the risk of exertional heat illness (heat stroke), which is life-threatening. Excess water loss through sweating (and the lungs, kidneys, and intestines) can reduce blood (plasma) levels (hypovolemia), and long periods under this condition can impair the brain and nervous system, cardiovascular function, and other factors that can

INFLAMMATION, PAIN, AND MEDICATION

Exercise-induced muscle damage from HIIT can lead to muscle soreness and pain, and many athletes use over-the-counter or prescription medications when symptomatic (193). While valid indications exist for nonsteroidal anti-inflammatory drug (NSAID) use in sports medicine, these drugs are not regulated (168). A recent review supports the potential for short-term NSAID use after an acute muscle injury (149), however, athletes may also have limited awareness of their effects and side effects (73). All pain medications have the potential for immediate and long-term side effects, including:

- The worsening of existing physical injuries (134).
- Delayed bone healing and muscle weakness (227).
- GI bleeding (even if not noticeable) and dysfunction (3, 231).
- Impaired tendon repair (232).
- Stroke (34).
- Low blood sodium (hyponatremia) (165).

Despite these risks, the high use of NSAIDs and other analgesics is widespread in all sports (46, 84, 226, 237). NSAID use in triathletes, for example, may be as high as 60% (73), with similar prevalence of NSAIDs and other analgesics in marathoners (107), and higher rates in both futsal and association football (172). Despite the lack of evidence on any added benefits of injected versus oral NSAID administration, Pedrinelli et al. (172) found that 15% of futsal (football) players injected NSAIDs. In addition to NSAIDs, other analgesics and muscle relaxants are also commonly used by athletes.

significantly reduce performance and health (221). As outlined in chapter 4, however, short-term periods of mild dehydration in the heat may offer an adaptive response in terms of plasma volume expansion and enhanced adaptation to heat (69).

A normal hydration status can be achieved with a range of total water intake (173). However, individual differences in sweat rates can lead to an array of conditions from dehydration to overhydration (47). In a study examining 2,135 endurance athletes who completed 42 km (26 mi) marathons, 109 km (67.8 mi) cycling events, and 226 km (140.4 mi) Ironman triathlons, about 60% of the athletes were dehydrated whereas 11% were overhydrated, 6% had mild hyponatremia (abnormally low blood sodium, <135 mEq/L), and 1% had severe hyponatremia (160). Excess water intake is even possible in team sport athletes (200). While reduced performance has been reported to occur with dehydration levels in excess of 3% to 5% or more of body weight, performance in the heat is maintained at controlled

and blinded dehydration levels up to 3% of body mass (37, 235), so despite its large exposure in the press, hydration status, at least in terms of dehydration, is typically overblown as an issue in the world of elite sport (47). Indeed, Sharwood et al. (199) showed no increased risk of heat illness associated with high levels of dehydration during the 2000 and 2001 South African Ironman triathlon, and high levels of weight loss did not significantly impair performance.

To help guide athletes performing HIIT in maintaining hydration balance—avoiding excess dehydration and overhydration—consider these suggestions:

- Have fluid available and encourage athletes to drink to thirst and not beyond (47).
- Those with a history of or suspected hydration problem can seek help from a clinician to test urine osmolality, specific gravity, and plasma sodium concentrations.

- Monitor body weight before and after training and competition. In general, a loss of body weight of 1 kg represents approximately 1 L of sweat loss (221), but remember that this is a natural occurrence and the body tolerates this well (47). Hypothetically, this could be countered by drinking 0.4 to 0.8 L/h of fluid (192).
- Body weight gain after training or competition may indicate excess fluid intake or overhydration, with the potential risk of hyponatremia (47).
- For daily self-assessment of hydration status, athletes can observe urine color, with pale (versus darker) yellow urine a general indicator of normal hydration (173). Achieving a pale urine color 2 to 4 h before training may be accomplished by consuming 5 to 10 mL/kg body weight (2 to 4 mL/lb), allowing time to void excess urine (192).
- The addition of sodium (salt) to food and water—to taste—can help fluid retention and improve hydration balance (189, 192).
- Rehydration following HIIT sessions or competition is important. The coingestion of food and water may work as well as sports drinks or milk-based beverages (14, 31).

Popular sports hydration recommendations are frequently influenced by commercial interests, including sports drink companies, sponsoring research (157).

Micronutrients

All vitamins and minerals make up the category of micronutrients and are essential for training, recovery and adaptation, competitive performance, and overall health. For example, various micronutrients are required in most aspects of body functions, playing vital roles in the conversion of macronutrients to energy (ATP), inflammatory regulation, immune function, muscle contraction and relaxation, brain function, hydration, and more.

The most important source of micronutrients is a healthy diet. However, athletes who eat a poor-quality diet or restrict calories from macronutrients may have suboptimal levels of micronutrients (221). Many athletes take dietary supplements, believing that they can improve athletic performance, but with some exceptions this idea is not supported in the scientific literature, and, in fact, some supplements can impair health (148, 171, 220). Like other nutri-

ents, an athlete's individual needs best dictate the use of a dietary supplement, and supplements should not replace a healthy diet. Blood tests for vitamins and minerals are commonly employed; however, many micronutrients don't necessarily reflect levels in muscles and other tissues (145) or reflect low acute dietary intake as indicated by dietary analysis (122).

Four key vitamins are highlighted here because they are commonly low in athletes (221).

Vitamin D

Technically a secosteroid hormone, vitamin D plays a key role in muscle function and recovery from training (164), can increase type 2 fast-twitch muscle fiber size, and may improve athletic performance (32). Related to its effect on bone health through improved calcium absorption and metabolism, vitamin D can help reduce the risk of stress fractures (188) and is important for immune, hormone, and inflammatory regulation (101). Vitamin D *insufficiency* is defined as blood levels of 20 to 30 ng/mL (50 to 75 nmol/L) with *deficiency* <20 ng/mL (<50 nmol/L) (221). However, peak athletic performance may be associated with levels of 50 ng/mL and above (32). Blood levels of vitamin D >100 ng/mL are considered an excess, with levels >150 ng/mL considered toxic (5).

Unfortunately, low levels of vitamin D are common in athletes and are associated with poor recovery, muscle pain and weakness, muscle atrophy, impaired power output, reduced coordination, increased risk for physical injury, and overall reduced exercise performance (32, 110, 201). Up to 94% of athletes in indoor sports and up to 80% in outdoor sports may have abnormally low vitamin D (201). As such, it is important for clinicians to routinely assess blood levels, especially in those with a history of poor vitamin D status. Athletes with darker skin (natural or tanned) and those who use sunscreen, wear protective clothing, have excess body fat, avoid the sun, or live at latitudes >35th parallel or who primarily train and compete indoors may be at higher risk for inadequate vitamin D (201).

The most common cause of poor vitamin D status is reduced sun exposure (110). A healthy range for adults and children of 50 to 80 ng/mL (125 to 200 nM/L) year-round appears adequate, with lower levels in this range found in late winter and early spring and higher levels in late summer (in the northern hemisphere). Measuring the 25(OH)D form of vitamin D versus the 1,25(OH)₂D form is a better indicator of vitamin D status and is the standard blood measurement used in clinical trials (90).

Vitamin D is produced by the conversion of 7-dehydrocholesterol in the skin following exposure to ultraviolet B radiation from the sun, the most readily available source of vitamin D for human metabolism (163). Peak skin production following a ~10 min exposure of summertime sun between 10 a.m. and 3 p.m., for example, can produce about 10,000 IU of vitamin D (201).

The most naturally occurring active form of this vitamin produced in the body from sun exposure is D3, or cholecalciferol, and is also found in fish liver oil, while the plant source D2, or ergocalciferol, is the form usually found in the diet (9). Foods provide only small amounts of vitamin D2, mostly through the fortification of junk foods. The use of vitamin D3 fish oil supplements may be necessary to restore and maintain normal vitamin D levels in athletes, with 10,000 IU per day defined as the tolerable upper-intake level without toxicity (reviewed by Jerome et al. (90)). However, deficiencies may require 50,000 IU of vitamin D3 per week for 8 wk under the guidance of a clinician (82).

Iron

A deficiency of iron, which may or may not cause anemia, can impair muscle function and athletic performance and produce symptoms of weakness and fatigue (121). Low iron levels can result from low intake of heme food sources, especially meat (18), with increased iron loss associated with high-altitude training, blood loss (such as gastric or menstrual), rapid growth (such as in adolescence), and, for some athletes, intense training due to iron loss in sweat, urine, feces, and from intravascular hemolysis (221). Low levels of riboflavin (vitamin B2) can also impair iron utilization (178).

Athletes who suspect low iron should visit a clinician who can perform a history, physical exam, and appropriate blood tests, which usually includes serum iron, ferritin (the stored form of iron), and total iron-binding capacity (transferrin) (154).

The use of lower-dose iron-containing dietary supplements as low as 10 mg can be useful to treat a diagnosis of iron deficiency. However, high doses of iron, such as 50 or 100 mg, can increase the risk of iron overload, observed as elevated ferritin that can increase oxidative stress and impair mitochondrial function (211).

Antioxidants

As noted earlier, dietary antioxidants play a key role in regulating inflammatory and immune responses

and recovery of high-intensity training, in great part due to their control of oxygen free radicals (177, 239). A healthy diet of fresh fruits and vegetables with meat, fish, and eggs can provide many natural antioxidants, not just vitamins and minerals but various phytonutrients, along with the amino acids cysteine and taurine (140, 177, 239).

However, dietary supplements containing antioxidants, in particular vitamins C and E and the phytonutrient beta-carotene, can become unwanted *prooxidants* and impair mitochondrial function and worsen oxidative stress (39, 140, 203, 248). These dietary supplements can also prevent training increases in $\dot{V}O_{2\max}$ (205).

Folate

As one of the B vitamins found in food, folate plays a key role in the synthesis of DNA, the metabolism of amino acids, red blood cell function (carrying oxygen to muscles), inflammation and oxidative stress regulation, and the prevention of megaloblastic anemia and insulin resistance (1, 179, 250). While blood tests for *folic acid*, an inactive form of folate, may appear normal, high numbers of people have a genetic mutation (called the C677T polymorphism of 5,10-methylenetetrahydrofolate reductase [MTHFR] gene) that impairs the conversion of folic acid (found in most dietary supplements and fortified processed foods) to the active form, folate (83). The MTHFR C677T and similar polymorphisms can be evaluated in a blood test, with folate levels measured in serum and as RBC folate (250). Athletes with a low folate status or the mutation should consider consuming more folate-rich foods (see table 7.4; (49)), avoiding inactive folic acid from dietary supplements (labeled as “folic acid”) and processed or fortified foods, including many energy bars and sports drinks, and, under the direction of a clinician, taking the active form (5-methyltetrahydrofolate, or 5-MTHF) as a dietary supplement to maintain normal blood levels. Adequate folate status could be achieved by eating a greater amount of natural folate-rich foods.

Body Composition

Food is sometimes used by athletes in an attempt to manipulate body composition: body fat and lean muscle mass. Many athletes focus on losing weight by reducing total calories. Deficient energy intake of one, two, or all three macronutrients, which can also reduce micronutrient intake, can impair

Table 7.4 Single-Serving Folate-Rich Foods

While there is no scientific definition of serving size, it is generally considered the amount of food customarily or reasonably consumed at one meal or a typical portion. Examples include 1/2 cup cooked vegetables, 1 cup raw vegetables or cooked whole grains, or 3 ounces of meat or fish.

Food	Naturally occurring folate (in mcg)
Turkey	486
Lentils	358
Spinach	263
Asparagus	243
Beef	221
Broccoli	168
Brussels sprouts	157
Beets	136
Leaf lettuce	119
Avocado	118
Papaya	112
Green peas	94
Orange	54

neuroendocrine and cardiometabolic health and lead to poor hormone, bone, immune, protein synthesis, and cardiovascular function (217). This problem, termed *relative energy deficiency in sport*, is a serious health risk in male and female athletes of all sports and ages, and one that can reduce performance (150).

Reducing calories is based on the idea of “calories in, calories out” and interpreted as eat less food and train longer or harder. However, this oversimplified concept is often not successful in long-term weight loss because the metabolism and endocrine response to food is not considered (8). For example, just reducing food calories can lower energy expenditure, increasing fat storage. A more sensible approach is to reduce the unwanted calories of junk food, such as cola or a candy bar, with its metabolic stress response. However, avoid reducing calories of nutrient-dense foods such as vegetables, healthy fats, or proteins. In addition, macronutrient composition influences metabolism and energy expenditure, as protein intake has a much greater thermic effect than carbohydrates or fat (78). Dietary fat is commonly restricted, due to its high caloric content, which can reduce the intake of the fat-soluble vitamins A, E, and K and essential fatty acids (88).

Over the course of a season, especially building up to HIIT with related higher energy expenditure, balanced body composition is important. Whether specifically pursued or naturally occurring, healthy

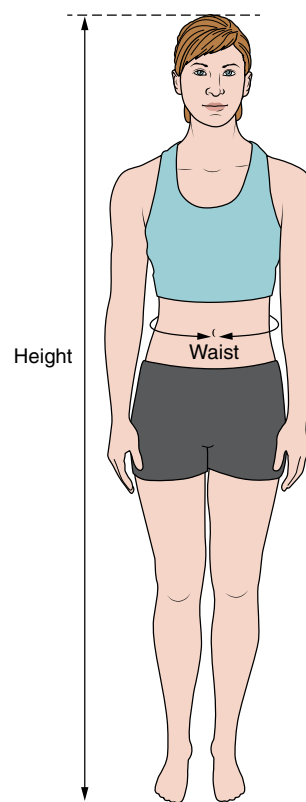


Figure 7.10 The waist-to-height ratio (WHtR) should be <0.5, meaning that the waist should be less than half the height.

changes in lean muscle mass and body fat, with potential weight adjustments, based on the specific needs of each athlete, are possible. Increased lean mass from added muscle can be valuable in sports such as American football (chapter 20), rugby (chapters 28 and 29), and ice hockey (chapter 25), while soccer players (chapter 30), runners (chapters 13 and 14), cyclists (chapter 15), and triathletes (chapter 19), as well as those in weight categories (i.e., combat sports and rowing, chapters 11 and 16, respectively) and aesthetic sports may require reduced weight, especially the loss of excess body fat, for better economy and/or meeting event requirements.

The term *overfat* refers to increased body fat that can impair health (126, 127). While the prevalence of excess body fat in athletes appears to be increasing (55, 244, 246), reducing it can also improve fitness, such as run performance (42, 185).

Measuring fat through body composition evaluation is commonly done with the body mass index (BMI), defined as weight in kg divided by height in m². However, this is not an accurate method that distinguishes lean muscle mass from body fat, with BMI misdiagnosing high numbers of people (127). The use of dual x-ray absorptiometry (DXA) may be the most effective method to accurately monitor body composition in athletes, including percent body fat (214). However, DXA can be costly for regular monitoring and not always accessible.

While there is no universally applicable criterion for body composition assessment (2), skinfolds and DXA are commonly used methods to monitor changes in body compositions (214). The use of skinfold thickness is commonly used to estimate percent body fat, with the American College of Sports Medicine recommending the use of Jackson and Pollock body density equations (142). There are limitations in this approach as fat distribution, subcutaneously and internally, is not similar for all individuals, varying across gender, ethnicity, body type, and age (71, 89).

A practical, simple, and inexpensive indirect measure of body fat that can be performed monthly by athletes, coaches, and clinicians is the waist-to-height ratio (WHtR). This highly accurate risk assessment for being overfat is useful for all ethnicities and ages in adults and children (11, 190). WHtR won't provide body fat percentage, which is without a consensus of normal cutoffs, but comes with a simple recommendation: the waist (measured at the level

of the umbilicus) should be less than half one's height, with a healthy ratio below 0.5 (127). This is particularly important for excess abdominal fat, which promotes body-wide low-grade chronic inflammation (60). Assessment of the WHtR is illustrated in figure 7.10.

Disordered Eating

In their extreme desire to change body composition, some athletes may develop unhealthy eating behaviors. Unhealthy eating behavior in male and female athletes, triggered by the desire to change body composition, can adversely affect HIIT due to deficits of energy, antioxidants, fatty acids, protein, and other nutrients. Disordered eating may be common in sports, appearing as a spectrum (53). The *Diagnostic and Statistical Manual of Mental Disorders* characterizes one extreme of the disorder as a gross disturbance in eating behavior that includes anorexia nervosa and bulimia nervosa, with higher numbers exhibiting less-severe but still unhealthy pathogenic eating behaviors (53). Features of the condition may include calorie restriction, fasting, and the use of laxatives, diuretics, diet pills, and vomiting. The problem appears more prevalent in sports requiring a minimum weight, although endurance athletes may mistakenly believe just weighing less will improve performance. Disordered eating may be more prevalent during competitive seasons or exist year-round, with athletes focusing on all foods or only part of the diet such as avoiding fat or protein. Addressing the problem starts with its recognition on the part of athlete, coach, and clinician, and sometimes a mental health practitioner.

Summary

HIIT offers many health and fitness benefits to athletes and can lead to improved performance. However, the combination of excess training stress causing neuroendocrine imbalance and the consumption of a poor diet, one high in sugar, can promote inflammation, metabolic substrate imbalance (reduced fat oxidation), and maladaptation, and contribute to physical, biochemical, and mental-emotional injuries. Athletes, coaches, and clinicians are encouraged to carefully monitor training progress (see chapters 8 and 9) and rely on healthy eating to maximize HIIT.

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